Appendix A Survey Template & Persona Data

Baseline experiment survey template recreated from Oswald et al. [34].

```
"What is the highest level of school you have completed or the highest
   degree you have received? <>"
"How much time do you spend online (browsing the web, using social media,
   etc.)? <>"
"How often do you use social media, including Reddit? <>"
"How often do you write comments online (on social media, in news outlets'
   comment sections, etc.)? <>"
"In politics people often talk about the 'left' and the 'right'. On a scale
    between 1 (furthest left) and 11 (furthest right), where would you
   place yourself? (scale horizontal) <>"
"How interested in politics are you? <>"
"How would you rate the Democratic Party? 0(negative) - 100(positive) <>"
"How would you rate the Republican Party? 0(negative) - 100(positive) <>"
"How much do you agree with the following statements?"
    "The government should not forgive student loan debt. <>"
    "Airbnb should be banned in cities. <>"
    "The federal minimum wage should be increased. <>"
    "The US should provide financial and military aid to Ukraine. <>"
    "A universal basic income would kill the economy. <>"
    "Climate change is one of the greatest threats to humanity. <>"
    "Fur clothing should be banned. <>"
    "The government should not invest in renewable energy. <>"
    "The US should provide humanitarian aid to Gaza and call for ceasefire.
       <>"
    "There should only be vegetarian food in cantines. <>"
    "Things like gender-neutral language and stating pronouns are silly
       issues. <>"
    "Prostitution should be illegal. <>"
    "Employers should mandate vaccination. <>"
    "The government should not be responsible for providing universal
       health care. <>"
    "Immigrants should be mandated to adopt the local language and culture.
       <>"
    "We need stricter gun control laws. <>"
    "The death penalty should be reestablished. <>"
    "Police officers should be wearing body cameras. <>"
    "Artificial Intelligence should replace humans where possible. <>"
    "Social media is a threat to democracy. <>"
"How much do you know about the following issues?"
    "Student Loan debt <>"
    "Airbnb <>"
    "Minimum wage <>"
    "War in Ukraine <>"
    "Universal basic income <>"
    "Climate change <>"
    "Fur clothing <>"
    "Renewable Energy <>"
```

```
    "Middle East <>"
    "Vegetarianism <>"
    "Gender neutral language <>"
    "Prostitution <>"
    "Vaccine mandates <>"
    "Universal health care <>"
    "Assimilation <>"
    "Gun control <>"
    "Death penalty <>"
    "Police body cameras <>"
    "Artificial Intelligence <>"
    "Social Media <>"
"To what extent do you agree with the following statement?"
    "I have a hard time understanding many political issues. <>"
    "I know a great deal about politics. <>"
    "I am very well informed about current political events. <>"
"How much do you trust..."
    "politics <>"
    "media <>"
    "science <>"
    "people generally <>"
"To what extent do you agree with the following statement?"
    "Quite a few of the people running the government are crooked. <>"
```

Appendix B Available Agent Actions

```
"refresh"                   - refreshes the paltform environment
"unlike_post"               - undo a previously given post like
"dislike_post"              - dislikes a given post
"undo_dislike_post"         - undo a previously given post dislike
"sign_up"                   - creates a profile on the platform (not callable by agens)
"create_comment"            - create a fist-level comment
"like_comment"              - like a given comment
"unlike_comment"            - undo a previously given comment like
"dislike_comment"           - dislike a given comment
"undo_dislike_comment"      - undo a previously given comment dislike
"create_comment_comment"    - create a 2nd-, 3rd-,... level comment
"continue_browsing"         - look for other content instead of engaging
```

Appendix C Activation Function

C.1 Base Activation Probability

Table C1 shows $P(X \geq k)$ for $k = 1, 2, 3$ over 24 time steps given specific values for pB. The values for pB were used in the activation function to model an agent's probability of activation given no prior activation in the thread simulation.

C.2 Burst Probability Modifier

As shown in Figure C1, the baseline experiment's data on IET (log-transformed) was of bimodal shape. According to this, I split the baseline data into two as shown by the line in Figure C1. The result was

Table C1: Probability of at least one, two, three activations over 24 simulation steps given different base activation probabilities according to the binomial probability function. An agent based on a seed user with self-reported behavior of commenting online multiple times a day was assigned $pB = 0.1$ and had a probability of 0.92 of being activated at least once over 24 steps.

Self-Reported Commenting-Behavior	pB	$P(X \geq 1)$	$P(X \geq 2)$	$P(X \geq 3)$
Never	.0025	.058	.002	–
About once per month	.00325	.075	.003	–
About once per week	.015	.304	.05	.005
About once per day	.05	.708	.339	.116
Multiple times a day	.1	.92	.708	.436

two different distributions, one for the IET within-bursts and the other for the IET between different bursts. Figure C2 shows this, presenting log-scaled histograms of the within-burst – i.e., intra-burst – and between-burst – i.e., inter-burst – IET data.

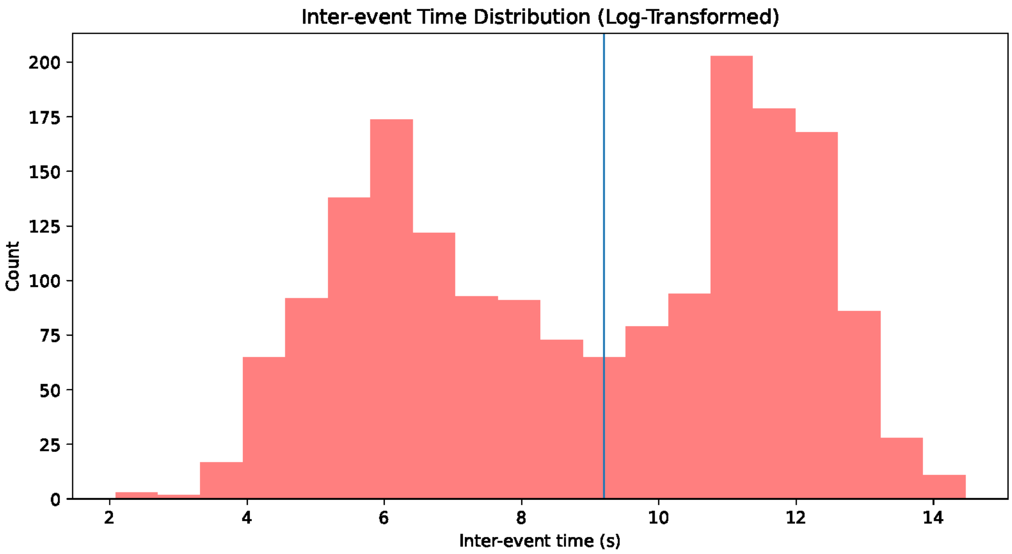

Fig. C1: Distribution of the log-transformed baseline IET data. The two modes represent within-burst and between-burst IET. Dividing the distribution at the vertical line returns the two individual distributions, as shown in Figure C2.

Using the `powerlaw` Python package [2], different distributions were fit to the data: a power-law, log-normal, and exponential distribution. As research has shown all three to be viable options in the modelling of online communication IET [10, 27, 30, 51, 56], all three were tested for model fit. Figure C3 shows QQ-plots of all three distributions compared to the intra-burst IET distribution of the baseline data. It shows that the log-normal distribution fits the baseline IET data best. The heavy tail indicates the transition from intra- to inter-burst.

Figure C4 supports the log-normal distribution as best fit for the data by plotting the probability density function (PDF), cumulative density function (CDF), and complementary cumulative density function (CCDF) of the baseline, log-normal, and power-law distributions. The log-normal fit clearly captures the intra-burst IET distribution of the baseline data best. Figure C5 shows intra-burst IET

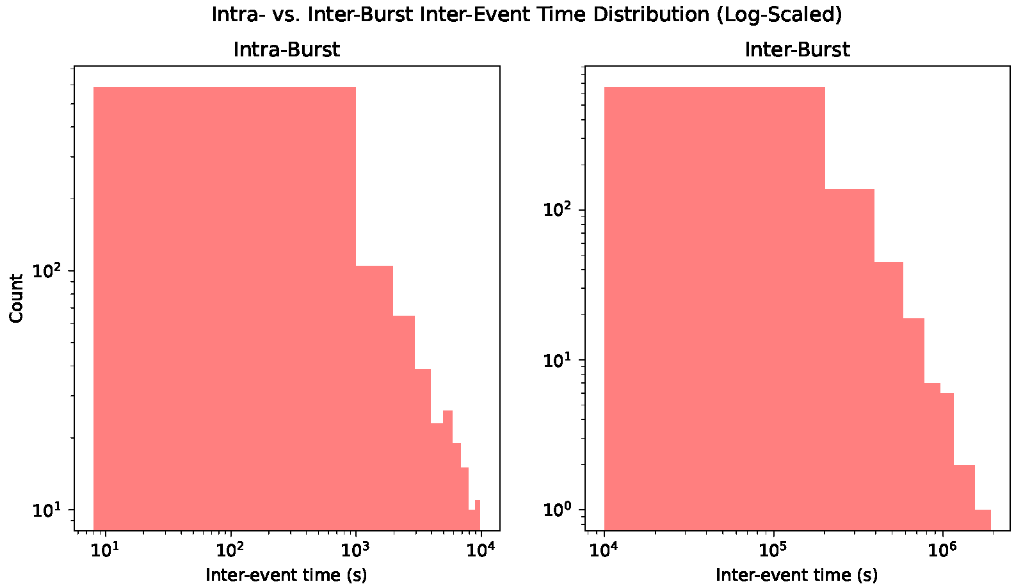

Fig. C2: Within-burst – i.e., intra-burst – and between-burst – i.e., inter-burst – IET distributions.

alongside inter-burst IET and the respective log-normal fits, along with a log-normal fit of the combined baseline IET distribution.

This log-normal fit was used to model the probability burst modifier pM in the activation function. Plugging the elapsed time (in seconds) since a user's last activation into the CCDF of the log-normal fit and using the resulting output as pM to bolster the probability of an agent being activated.

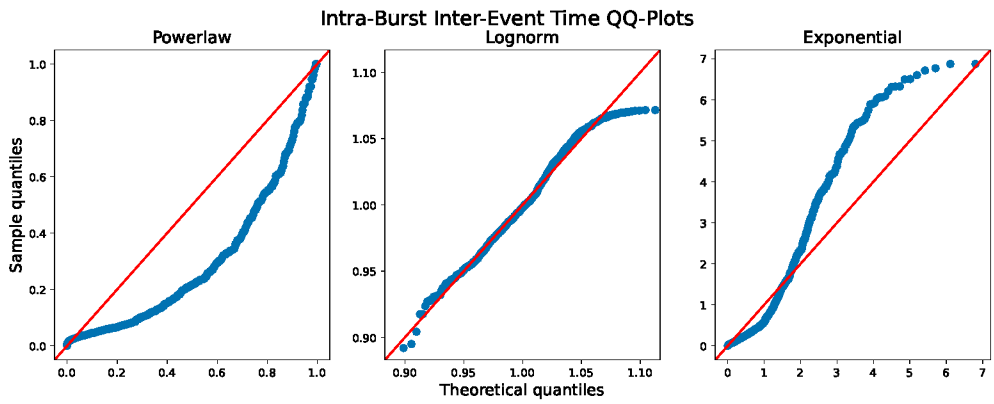

Fig. C3: QQ-plots for the powerlaw, log-normal, and exponential distribution fits. The three different distributions' quantiles are plotted against the baseline IET data's quantiles. Visual analysis suggests the log-normal fit as the best fit for the data.

Appendix D Further Results

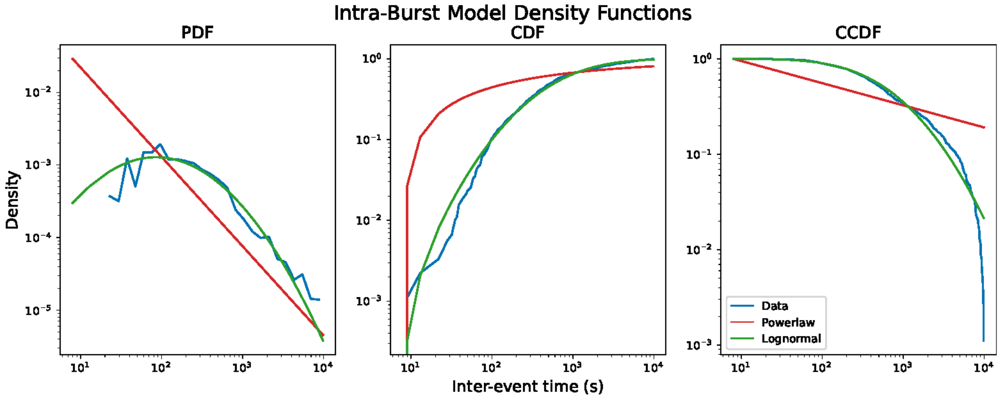

Fig. C4: PDF, CDF, and CCDF plot of the powerlaw and log-normal fit. The individual plots show the log-normal fit to better model the baseline data.

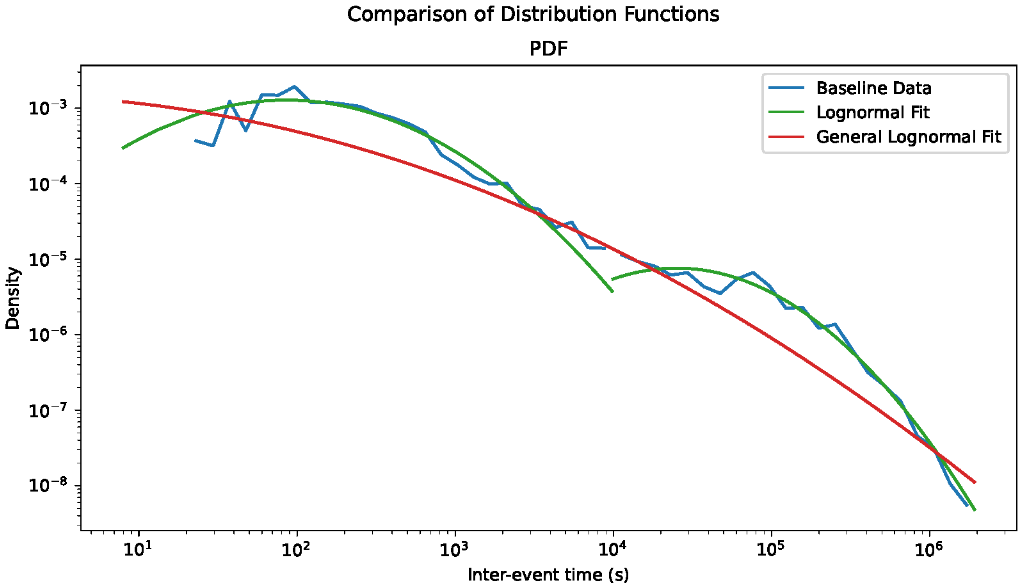

Fig. C5: Combined PDF plot of intra- and inter-burst IET. The blue and green lines show the intra- and inter-burst IET of the baseline data and log-normal fit individually. The red line shows the log-normal fit of overall IET.

Table D2: Overview of the number of active users/agents, comment volume, reply count, and average toxicity for each thread. The simulated values are averaged over 10 runs and the standard deviation (SD) is given in parentheses.

Thread	Active		Volume		Reply Count		Toxicity	
	Baseline	Sim (SD)	Baseline	Sim (SD)	Baseline	Sim (SD)	Baseline	Sim (SD)
A universal basic income would kill the economy.	32	107 (5)	51	272 (18)	21	11 (3)	0.077	0.139 (0.099)
Airbnb should be banned in cities.	50	43 (4)	95	119 (14)	9	8 (1)	0.042	0.099 (0.099)
Artificial intelligence should replace humans where possible.	56	25 (1)	118	95 (11)	29	14 (3)	0.037	0.090 (0.087)

Table D2: Overview of the number of active users/agents, comment volume, reply count, and average toxicity for each thread. The simulated values are averaged over 10 runs and the standard deviation (SD) is given in parentheses.

Thread	Active		Volume		Reply Count		Toxicity	
Climate change is one of the greatest threats to humanity.	39	94 (5)	51	245 (13)	16	10 (3)	0.060	0.117 (0.110)
Employers should mandate vaccination.	36	63 (4)	90	197 (24)	36	12 (4)	0.069	0.110 (0.116)
Fur clothing should be banned.	38	44 (3)	63	135 (16)	12	9 (4)	0.082	0.159 (0.124)
Immigration should be regulated more strictly.	38	72 (4)	46	225 (17)	5	11 (2)	0.079	0.170 (0.116)

Table D2: Overview of the number of active users/agents, comment volume, reply count, and average toxicity for each thread. The simulated values are averaged over 10 runs and the standard deviation (SD) is given in parentheses.

Thread	Active		Volume		Reply Count		Toxicity	
Police officers should wear body cameras.	55	104 (4)	89	254 (28)	7	11 (3)	0.079	0.111 (0.115)
Prostitution should be illegal.	33	30 (2)	59	117 (17)	18	14 (4)	0.172	0.220 (0.103)
Social media is a threat to democracy.	52	51 (3)	95	179 (14)	21	14 (3)	0.096	0.171 (0.128)
The US should condemn Israel's military actions in Gaza as acts of genocide and impose full sanctions.	34	103 (4)	94	265 (14)	25	11 (3)	0.181	0.212 (0.106)

Table D2: Overview of the number of active users/agents, comment volume, reply count, and average toxicity for each thread. The simulated values are averaged over 10 runs and the standard deviation (SD) is given in parentheses.

Thread	Active		Volume		Reply Count		Toxicity	
The US should provide financial and military aid to Ukraine.	45	105 (5)	83	290 (17)	10	10 (3)	0.135	0.190 (0.112)
The death penalty should be reestablished US-wide.	59	86 (5)	141	235 (14)	66	13 (4)	0.148	0.237 (0.095)
The federal minimum wage should be increased.	53	98 (6)	102	255 (18)	10	9 (3)	0.061	0.110 (0.106)
The government should not be responsible for providing universal health care.	42	86 (6)	75	249 (10)	10	10 (3)	0.076	0.167 (0.117)

Table D2: Overview of the number of active users/agents, comment volume, reply count, and average toxicity for each thread. The simulated values are averaged over 10 runs and the standard deviation (SD) is given in parentheses.

Thread	Active		Volume		Reply Count		Toxicity	
The government should not forgive student loan debt.	41	84 (4)	119	228 (20)	71	9 (3)	0.062	0.158 (0.117)
The government should not invest in renewable energy.	35	25 (4)	42	94 (18)	5	15 (6)	0.076	0.112 (0.105)
There should only be vegetarian food in cantines.	67	40 (6)	126	124 (22)	7	10 (3)	0.059	0.111 (0.114)
Things like gender-neutral language and stating pronouns are silly issues.	62	27 (5)	252	99 (28)	122	15 (5)	0.158	0.257 (0.129)

Table D2: Overview of the number of active users/agents, comment volume, reply count, and average toxicity for each thread. The simulated values are averaged over 10 runs and the standard deviation (SD) is given in parentheses.

Thread	Active		Volume		Reply Count		Toxicity	
We need stricter gun control laws.	44	92 (6)	106	223 (21)	43	9 (3)	0.139	0.157 (0.099)

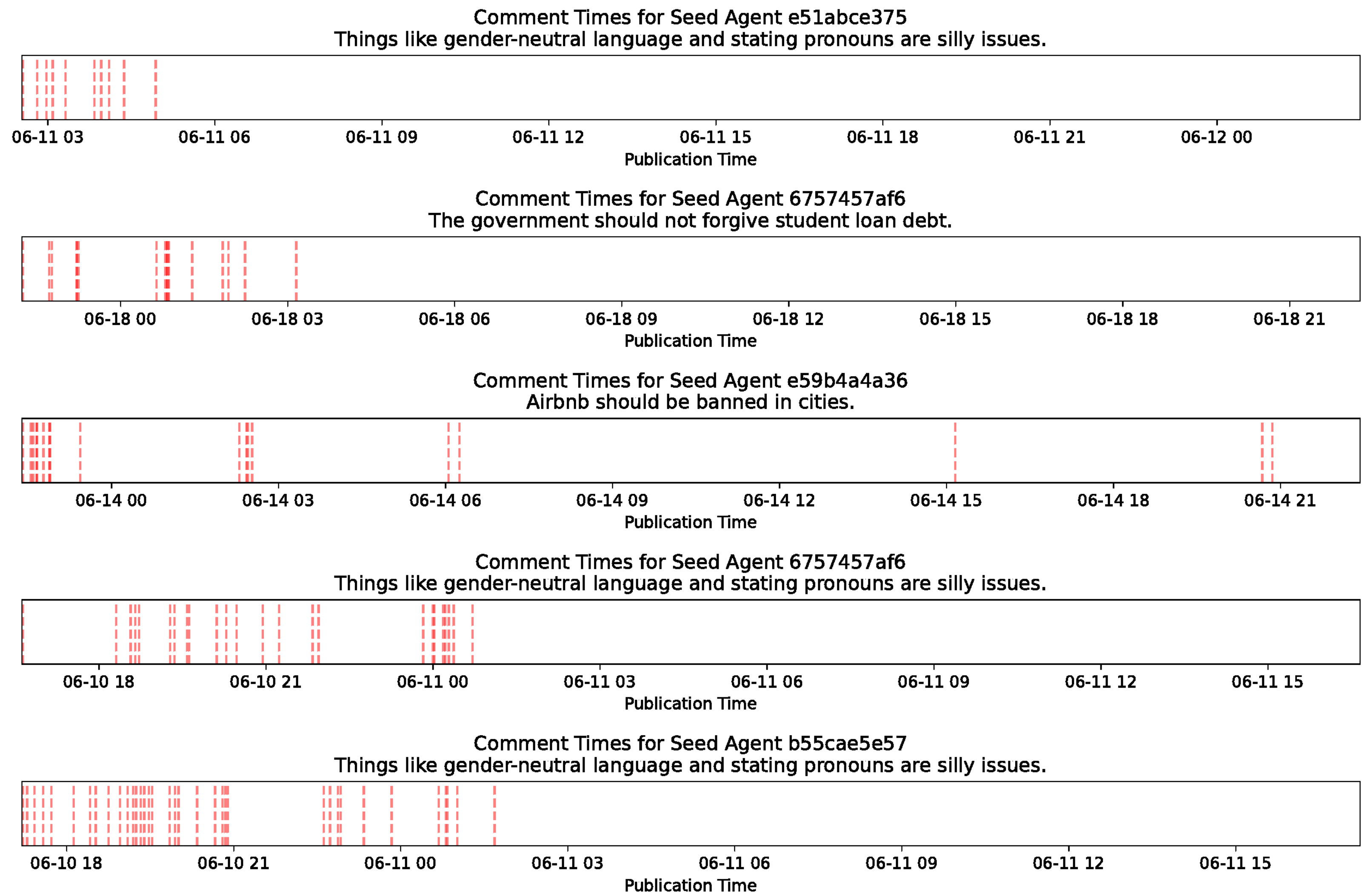

Fig. D6: Exact commenting times of the five most active seed users and their most commented on thread, visualizing interaction bursts followed by downtime.

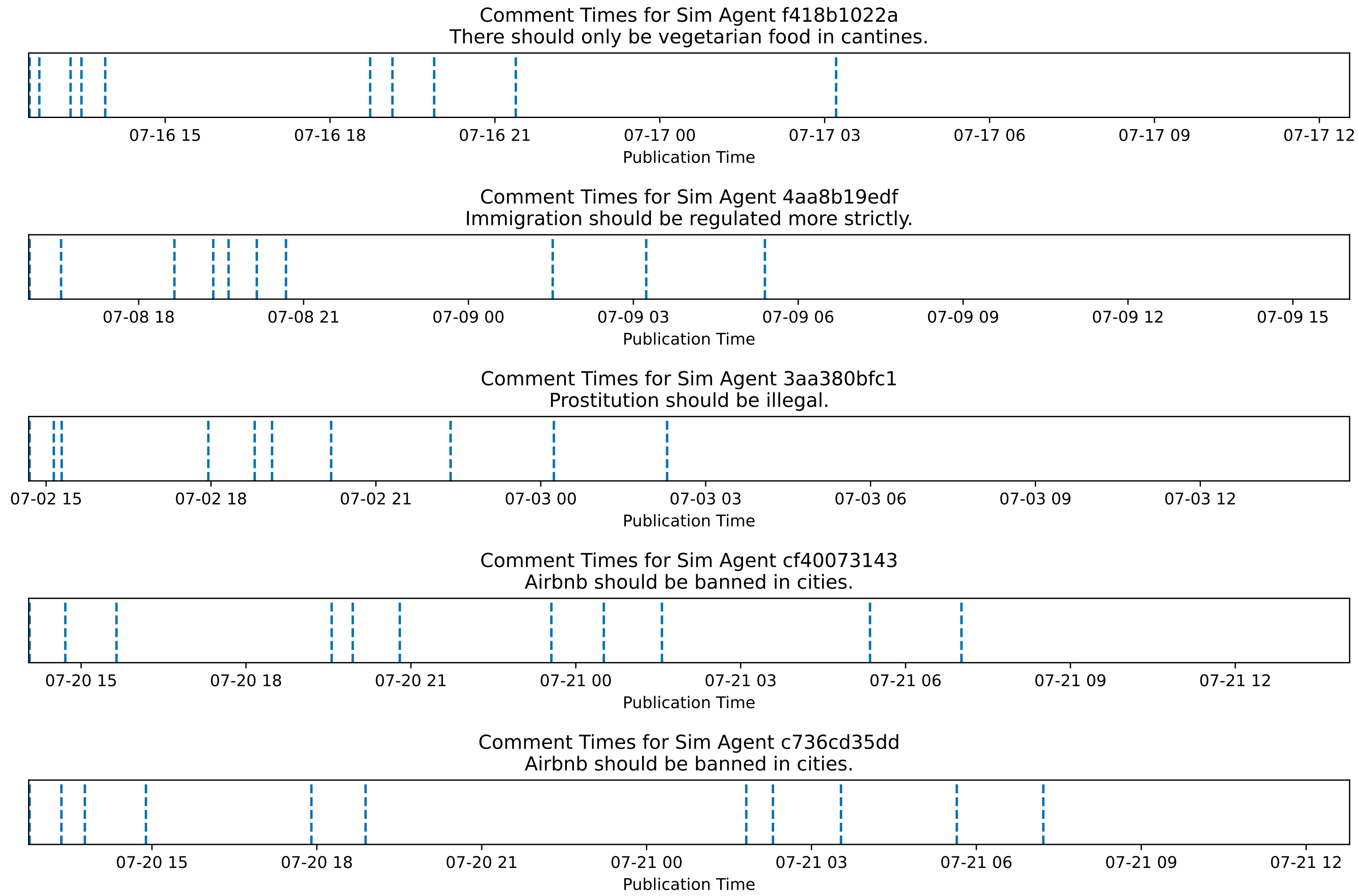

Fig. D7: Exact commenting times of the five most active simulation agents and their most commented on thread, visualizing interaction bursts followed by downtime.